A Low-Cost Multi-Robot Experimental Platform for Mobile Robotics Education and Research

J. Bautista
Faculty of Engineering
Systems Undergraduate Program
University of the Amazonia
Florencia, Caquetá, Colombia
jaid.bautista@udla.edu.co

M. A. Pastrana
Dept. of Mechanical Engineering
Mechatronics Graduate Program
University of Brasilia
Brasilia, DF, Brazil
mario.pastrana@ieee.org

Jesus Pinto-Lopera
Faculty of Engineering
Systems Undergraduate Program
University of the Amazonia
Florencia, Caquetá, Colombia
jes.pinto@udla.edu.co

Daniel M. Muñoz
Mechatronics Graduate Program and
Faculty of Gama, Electronics Eng.
University of Brasilia
Brasilia, DF, Brazil
damuz@unb.br

***Abstract*—This paper presents a low-cost multi-robot experimental platform for mobile robotics education and research. The system integrates three ESP32-based differential-drive robots, vision-based localization using OpenCV and ArUco markers, and a CoppeliaSim environment with custom CAD models. With an approximate total cost of USD 154, the platform provides an accessible alternative for institutions with limited robotic infrastructure. A PID-based trajectory-tracking module with support for bio-inspired gain tuning using Particle Swarm Optimization and Moth-Flame Optimization is also included. The platform was evaluated in two multi-robot navigation scenarios executed in both simulation and real environments. The results show that the robots were able to follow the planned trajectories and reach their target regions, while the simulator reproduced the general motion behavior of the physical system. Despite discrepancies caused by wheel slippage, open-loop actuation, localization noise, and unmodeled environmental effects, the proposed platform represents a practical tool for teaching and validating fundamental concepts in mobile robotics, including kinematics, trajectory planning, pose control, computer vision, and simulation-to-real experimentation.**

***Index Terms*—Low-cost robotics platform, mobile robotics, bio-inspired optimization, PID controller tuning, trajectory tracking**

I. Introduction

In recent years, mobile robots have gained increasing popularity due to their ability to operate autonomously in dynamic environments. They have been widely applied in precision agriculture for tasks such as crop monitoring and autonomous harvesting [1], as well as in delivery services for logistics and last-mile transportation [2]. Additionally, mobile robots have shown potential in several other areas, including industrial inspection, healthcare, and environmental monitoring [3]. However, the financial cost of building mobile robots, together with the technical knowledge required to perform experiments in mobile robotics, remains a significant challenge. As a result, the barrier to entry for experimental research and development in mobile robotics continues to increase [4].

In an effort to reduce these barriers and democratize access to mobile robotics, virtual and remote laboratories have emerged as accessible solutions for implementing and validating control algorithms in both simulated and real environments [5]. Early work by Greenwald and Kopena demonstrated that low-cost robotic kits could effectively support robotics education and research through practical activities such as motor control, dead reckoning, and map building [6].

Subsequent studies expanded this concept by integrating web-based platforms and remote experimentation. Lopes *et al.* developed a browser-based environment for programming and controlling mobile robots, simplifying access for students in robotics and computer science courses [7] . Similarly, Acero and Sanchez introduced TELEROBOT-LAB, a remote laboratory based on a web platform that enabled real-time control and monitoring of Pioneer P3-DX and AmigoBot robots, while also supporting collaborative observation of experiments [8]. Casini *et al.* proposed a remote laboratory using LEGO Mindstorms robots integrated with Matlab for decentralized control experiments [9], while Iturrate *et al.* combined serious games and remote robotics to improve STEM learning engagement [10].

Other contributions focused on the infrastructure and scalability of remote laboratories. Wilson and Egerstedt introduced the Robotarium, a remotely accessible multi-robot platform that enables researchers worldwide to validate multi-agent algorithms on real robotic systems [5]. Follow-up studies further improved the platform through automation, wireless charging, and robust safety mechanisms, allowing thousands of experiments to be executed with minimal human intervention [4], [11].

Table I summarizes the remote laboratory platforms discussed in this section. It is important to note that most of these laboratories have been developed outside Latin America, with the notable exceptions of the works by Acero and Sanchez in

Colombia and Lopes *et al.* in Brazil. In terms of hardware cost, several platforms rely on relatively expensive robotic systems. For example, the studies by Greenwald and Kopena and by Casini *et al.* use LEGO-based kits, which typically cost around USD 1,200, making the deployment of multi-robot laboratories considerably more expensive. In contrast, Wilson *et al.* [4] employed the GRITSBot X platform, with an estimated cost of USD 225 per robot. In the Latin American context, Acero and Sanchez used the Pioneer P3-DX, a platform that has been discontinued and currently costs approximately USD 1,000 on the second-hand market.

Despite substantial progress in remote and experimental robotics platforms, several limitations remain. First, few platforms provide complete open-source CAD models and simulation-ready files. Second, many available systems rely on moderately expensive robotic platforms, which limits their adoption in resource-constrained educational environments. Third, few low-cost educational platforms include controller-tuning modules that allow students and researchers to evaluate the effect of optimization techniques on trajectory-tracking performance. Finally, the literature still lacks low-cost and easily replicable multi-robot platforms specifically designed to support engineering education and experimental mobile robotics research in Latin America.

Motivated by these gaps in the literature, this paper presents a low-cost and easily replicable multi-robot experimental platform based on widely available hardware and open-source software. The proposed system uses differential-drive robots manufactured by KUONGSHUN, each costing approximately USD 40 and controlled by an ESP32 microcontroller. Complete three-dimensional CAD models and CoppeliaSim-compatible files are freely available in a public GitHub repository [12], enabling straightforward replication and customization. Robot position and orientation are estimated using a conventional Wi-Fi camera or smartphone together with Python and OpenCV-based ArUco marker detection [13]. In addition, the platform includes a PID-based trajectory-tracking module with support for bio-inspired gain tuning, allowing users to analyze the effect of controller parameters on real and simulated robot behavior. Beyond its educational purpose, the proposed platform is intended to provide a replicable experimental testbed for validating trajectory-tracking and multi-robot navigation algorithms under real-world constraints. Therefore, the platform is aligned not only with robotics education, but also with experimental mobile robotics research, particularly in simulation-to-real validation.

II. Differential-Drive Mobile Robot

Figure 1 illustrates the kinematic model of a wheeled mobile robot, featuring two coaxial rear driving wheels. In this model, P denotes the geometric center of the line connecting the left and right wheels, L represents the distance between the wheels, and d is the wheel diameter. The angular velocities of the left and right wheels are denoted by ω_l and ω_r, respectively. The robot's pose is defined by the vector $q = (x, y, \theta)^T$ in the global coordinate system XOY, where (x, y) are the coordinates of the center point P, and θ is the angle between the robot's direction of motion and the horizontal axis [14].

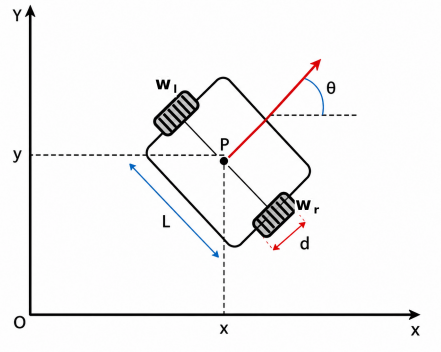


Fig. 1. Differential robot

To calculate the values of ω_l and ω_r based on the kinematic model, the differential drive robot equations are used as shown below:

$$\dot{x} = \frac{R}{2} \times (\omega_r + \omega_l) \times \cos\theta, \tag{1}$$

$$\dot{y} = \frac{R}{2} \times (\omega_r + \omega_l) \times \sin\theta, \tag{2}$$

$$\dot{\theta} = \frac{R}{L} \times (\omega_r - \omega_l), \tag{3}$$

where $R = \frac{d}{2}$, $\dot{x}$ represents the velocity in the X-direction, $\dot{y}$ represents the velocity in the Y-direction, and $\dot{\theta}$ is the angular velocity of the robot. Furthermore, the kinematic equations from [14] were used and are presented below:

$$\dot{x} = v \times \cos\theta, \tag{4}$$

$$\dot{y} = v \times \sin\theta, \tag{5}$$

$$\dot{\theta} = \omega, \tag{6}$$

where v is the linear velocity of the robot, and ω is the angular velocity. Using these equations, ω_l and ω_r are calculated and presented below:

$$\omega_l = (2v - \omega L)/d, \tag{7}$$

$$\omega_r = (2v + \omega L)/d. \tag{8}$$

TABLE I
SUMMARY OF METHODOLOGIES AND KEY PERFORMANCE INDICATORS FOR REMOTE ROBOTICS LABS

Authors	Year	Country	Methodology	Principal Points	Metrics Comparison
Greenwald & Kopena	2003	USA	Detailed lab assignments using low-cost Handy Board/LEGO kits and Interactive C.	Pushing low-cost hardware for research/educational goals like dead reckoning and map building.	Sensor sampling accuracy, odometry effectiveness, and build time.
Acero & Sanchez	2011	Colombia	TELEROBOT-LAB: Web platform using PHP and Java Applets for connection to Pioneer P3-DX and AmigoBot robots.	Concept of configurable test "scenarios"; features an emergency stop button for hardware integrity and an "Observer" role for demos.	Usability (SUS-style survey), system response time perception, and video feedback accuracy.
Iturrate et al.	2013	Spain / Portugal	WebLab-Deusto with Serious Games and Google Blockly.	Gamification of experiments to increase engagement; labyrinth navigation to answer questions.	Student engagement (flow) and ease of use for logic principles.
Casini et al.	2014	Italy	LEGO Mindstorms; control laws coded in Matlab; Java-based web interface.	Versatile for teaching and research; virtual obstacles and simulated sensors for motion coordination.	Multi-agent coordination success and platform independence.
Lopes et al.	2017	Brazil	Control and Programming Environment (CPE); remote lab, online programming tool, and virtual learning environment.	Teaching computer science through mobile robots; C/C++ or graphical programming via web browser.	Student academic scores, interest levels, and task completion
Wilson *et al.*	2020	USA	User-centric evolution study reporting on the global impact and demographics of the user base.	Highlights challenges of providing access to inexperienced users and how feedback guides standardized testbed enhancements.	Diversity of use cases (e.g., traffic modeling, biomimicry) and demographic range of the user base.
Wilson *et al.*	2021	USA	Full system automation through hardware development (GRITSBot X) and robust Control Barrier Functions (CBFs).	Reduces human intervention through physical robustness, automated wireless charging, and safety protocols against disturbances.	System reliability (intervention needed for only 0.76% of experiments) and hardware maintenance efficiency.
Wilson & Egerstedt	2023	USA	Robotarium: Large-scale testbed using Docker containers.	Democratized research; high-fidelity hardware facility to validate algorithms in the real world.	Usage trends, robot pose accuracy, and execution time delays.
This paper	2026	Colombia / Brazil	Low-cost multi-robot platform integrating CoppeliaSim, OpenCV/ArUco localization, ESP32 differential-drive robots, and bio-inspired PID tuning.	Replicable platform for mobile robotics education and experimental validation in Latin America.	Trajectory error and implementation cost.

III. PROPOSED METHODOLOGY

A. System Architecture

The proposed methodology is illustrated in Fig. 2. As shown in the figure, the proposed platform is organized into two interconnected domains: the *software world* and the *hardware world*. In the software world, a host computer runs the CoppeliaSim environment, which contains the virtual models of the mobile robots, the trajectory generator, the PID controller, and the bio-inspired optimization algorithms. This environment is used to perform automatic controller tuning and to validate the control strategies before their deployment on the physical robots.

In the hardware world, three differential-drive robots operate within an experimental arena and are monitored by an overhead camera. The camera captures images of the workspace, and a Python application based on OpenCV estimates the position and orientation of each robot through ArUco marker detection. The estimated poses are transmitted to the host computer, where the control actions are computed in real time.

The host computer sends wheel commands in PWM format to the robots through a Wi-Fi connection using the UDP communication protocol. Since the robots are not equipped with wheel encoders, these commands are applied in open loop at the motor level. Nevertheless, the overall system operates as a closed-loop control architecture because the robot poses are continuously updated by the external vision-based localization system.

B. Low-Cost Mobile Robot Platform

As discussed in the Introduction, the robotic platform employed in this work is a commercially available differential-drive robot manufactured by KUONGSHUN, with an approximate cost of USD 40 per unit. Each robot is equipped with two DC motors, an H-bridge motor driver, a rechargeable lithium battery, and an ESP32 microcontroller responsible for wireless communication and motor actuation.

One of the main limitations of this platform is the absence of a simulation-ready model that can be directly imported into CoppeliaSim. To overcome this limitation, a complete three-dimensional CAD model of the robot was developed and subsequently exported to the simulation environment. This model accurately reproduces the geometry and dimensions of the physical platform, enabling high-fidelity simulations and facilitating direct transfer of control algorithms from simulation to the real robots.

Figure 3 presents the physical robot and its corresponding CAD model. All design files, including the CAD models, Cop-

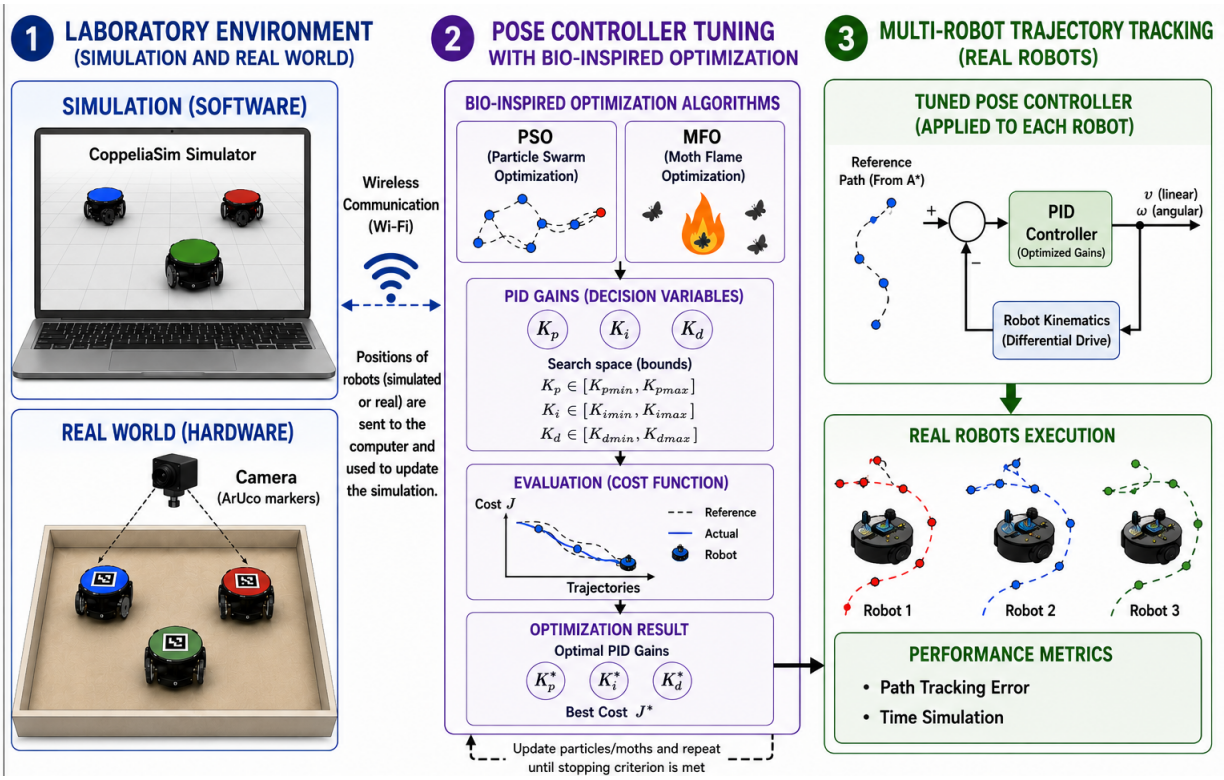
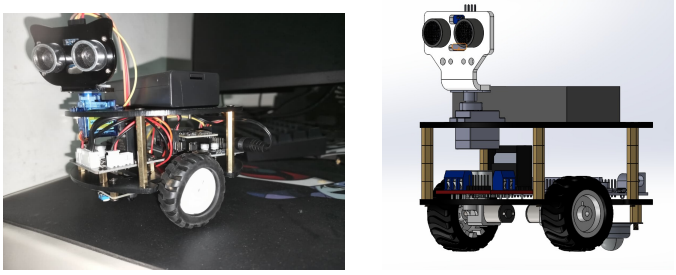


Fig. 2. General architecture of the proposed multi-robot experimental platform.



(a) Physical robot.

(b) CAD model developed in SolidWorks.

Fig. 3. Low-cost differential-drive robot manufactured by KUONGSHUN and its corresponding three-dimensional CAD model.

peliaSim scenes, and control software, are freely available in a public GitHub repository [12]. This open-source contribution allows other researchers and educators to reproduce, modify, and extend the proposed platform for educational and research purposes.

C. PID-Based Pose Controller

PID controllers are widely used in mobile robotics due to their simplicity, low computational cost, and practical effectiveness in trajectory-tracking tasks. In general, the PID control law is expressed as

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt},$$

where $e(t)$ is the tracking error, and K_p, K_i, and K_d are the proportional, integral, and derivative gains, respectively.

In this work, the PID controller computes the linear and angular velocity commands required for each robot to follow a sequence of reference waypoints. The proper selection of the PID gains is therefore essential to reduce the trajectory-tracking error and improve the agreement between simulated and real robot behavior.

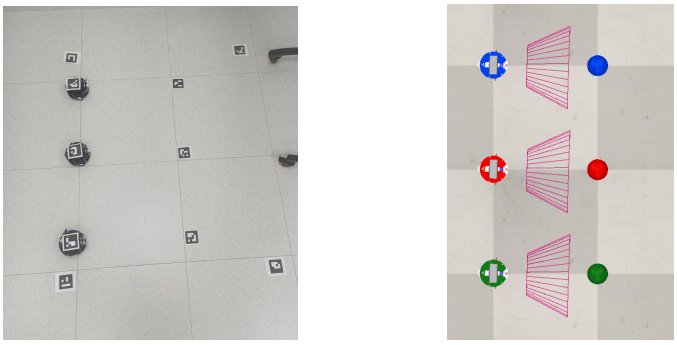
The proposed platform also supports automatic gain tuning using bio-inspired optimization algorithms, such as Particle Swarm Optimization (PSO) [15] and Moth-Flame Optimization (MFO) [16]. These methods are used to search for suitable PID gains in simulation before deploying the controller in the physical robots.

Figure 2 illustrates the trajectory-planning and tracking process. The optimized PID controller computes the linear and angular velocities required for each robot to follow the generated path accurately. In this work, three robots were experimentally validated by executing the same reference trajectory in both the simulation and hardware environments. The tracking errors observed in both domains were then compared to evaluate the fidelity of the simulation model and the effectiveness of the proposed control methodology.

The next section presents the experimental results obtained with the proposed platform, including the PID gain-tuning results and the trajectory-tracking performance achieved in both simulated and real-world experiments.

IV. Results and Discussion

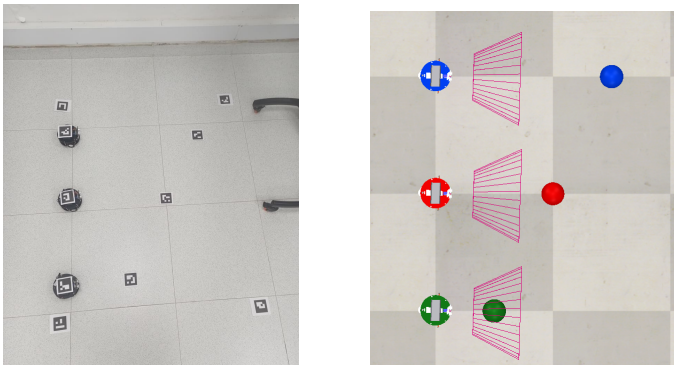
The total cost of the proposed platform was approximately USD 154, including three differential-drive robots (USD 40 each) and a vision-based localization camera (USD 34). This cost is substantially lower than that of commonly adopted educational and research platforms, such as LEGO Mindstorms and Pioneer P3-DX robots, whose costs typically exceed USD 1,000. Figures 4a and 4b illustrate the real and simulated scenario 1 used to evaluate the proposed low-cost mobile robotics platform. In Scenario 1 (Fig. 4a), the three robots are assigned target positions aligned along the same axis, requiring coordinated navigation toward their respective destinations.



(a) Real scenario 1. (b) Simulation scenario 1.

Fig. 4. Real and simulated environments for Scenario 1.

In contrast, Scenario 2 (Fig. 5) employs the same robotic configuration but modifies the target positions by introducing displacements along both the x- and y-axes. This second scenario increases the complexity of the navigation task and provides a more comprehensive assessment of the trajectory-tracking and trajectory-tracking capabilities of the multi-robot system.



(a) Real scenario 2. (b) Simulation scenario 2.

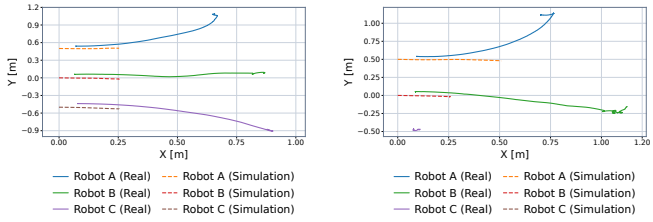
Fig. 5. Real and simulated environments for Scenario 2.

Before evaluating the trajectory-tracking performance, the PID gains were tuned in simulation using PSO and MFO algorithms. The objective function was defined as

$$J = \left(\frac{E_{acc}}{N_{iter}} \right) + e_d,$$

where E_{acc} represents the accumulated trajectory-tracking error, N_{iter} is the number of iterations, and e_d is the final distance error with respect to the target position. Each optimization algorithm was executed once as an initial tuning stage for the proposed platform. The gains obtained with PSO were $K_p = 0.3432$, $K_i = 0.0001$, and $K_d = 0.0001$, whereas the gains obtained with MFO were $K_p = 0.3413$, $K_i = 0.1230$, and $K_d = 0.0049$. These results are reported as representative tuning outcomes rather than a statistical comparison, since only one execution was performed for each algorithm.

The corresponding trajectory-tracking results obtained in simulation and real experiments are shown in Fig. 6 and the videos are hosted at the YouTube platform [17]. In both scenarios, the robots successfully reached their assigned target positions while preserving the overall navigation behavior predicted by the simulator. A general agreement between the real and simulated trajectories can be observed, indicating that the simulation environment reproduces the main motion tendencies of the robotic platform. Deviations are present throughout the trajectories, particularly during turning maneuvers and trajectory transitions, which can be attributed to wheel slippage, actuator uncertainties, localization inaccuracies, communication delays, and environmental factors that are not fully represented in the simulation model.



(a) Scenario 1. (b) Scenario 2.

Fig. 6. Real and simulated trajectory-tracking results for the evaluated multi-robot scenarios: (a) Scenario 1 and (b) Scenario 2.

To quantitatively evaluate the fidelity of the simulation environment, the Euclidean distance error between the real and simulated trajectories was computed after resampling both trajectories to the same number of points. Table II summarizes the obtained error metrics, including the mean error, RMSE, median error, and maximum observed deviation for each robot. In Scenario 1, all three robots exhibited comparable performance, with RMSE values ranging from 0.3609 m to 0.4456 m. The results indicate a consistent agreement between simulation and experimental data across the fleet, suggesting that the simulator is capable of reproducing the overall motion behavior of the robots under relatively simple navigation conditions.

Scenario 2 introduced a more demanding navigation task due to the modified target configuration. As a result, Robots A and B exhibited larger discrepancies between the simulated and experimental trajectories, reaching RMSE values of 0.5555 m and 0.5578 m, respectively. In contrast, Robot C achieved substantially lower errors, obtaining a mean tra-

TABLE II
TRAJECTORY ERROR BETWEEN REAL AND SIMULATED ROBOTS FOR THE TWO EVALUATED SCENARIOS.

Scenario	Robot	Mean [m]	RMSE [m]	Median [m]	Max [m]
Scenario 1	A	0.3426	0.4456	0.2650	0.7243
	B	0.2838	0.3609	0.1886	0.6247
	C	0.3195	0.4166	0.1835	0.7527
Scenario 2	A	0.4758	0.5555	0.6680	0.7777
	B	0.4486	0.5578	0.3734	0.8811
	C	0.1017	0.1021	0.1069	0.1174

jectory error of 0.1017 m and an RMSE of 0.1021 m. This result suggests that the accuracy of the simulation model may vary depending on the robot path geometry and the specific maneuvers executed during the experiment.

It is important to note that the proposed simulator is intended as a low-cost educational and preliminary validation environment rather than a high-fidelity dynamic model. The relatively large errors observed in some trajectories are consistent with the limitations of the physical platform, including open-loop motor actuation, the absence of wheel encoders, wheel slippage, and vision-based localization noise. Nevertheless, the obtained results show that the simulator can reproduce the general motion tendencies of the real robots, which is sufficient for teaching, controller prototyping, and initial simulation-to-real validation.

V. CONCLUSION

This paper presented a low-cost multi-robot experimental platform for mobile robotics education and research. The system integrates three ESP32-based differential-drive robots, vision-based localization using OpenCV and ArUco markers, and a CoppeliaSim environment with CAD models and control software released as open-source resources. With an approximate cost of USD 154, the platform offers an accessible alternative for teaching and experimental validation in resource-constrained institutions.

The platform was evaluated in two multi-robot navigation scenarios using both simulated and real robots. The results showed that the robots were able to follow the planned trajectories and reach their target regions, while the simulation reproduced the general motion behavior of the physical system. However, discrepancies between simulated and real trajectories were observed due to wheel slippage, open-loop motor actuation, localization noise, communication delays, and unmodeled environmental effects.

Overall, the proposed platform provides a practical tool for studying differential-drive kinematics, trajectory planning, pose control, computer vision, and simulation-to-real validation. Future work will focus on improving simulation fidelity, performing repeated experimental trials, adding multi-robot coordination metrics, and developing a web-based interface for remote experimentation.

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